

# MATH403: Homework 3

JAMES ZHANG<sup>\*</sup>

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10. Given that  $\{e, a, a^2, b, ab, a^2b\}$  is a group where  $|a| = 3$ ,  $|b| = 2$  and  $ba = a^2b$ , determine which of the six elements is  $aba^2$  and which is  $a^2bab$ .

*Proof.*

- (i) Let's start with  $aba^2$ . Note that  $aba^2 = a(ba)a$  by associativity of groups. We're given  $ba = a^2b$ . By direct substitution,

$$aba^2 = a(a^2b)a = a^3ba$$

Since  $|a| = 3 \implies a^3 = e$  by definition of order of an element. Thus,

$$a^3ba = eba = ba$$

Finally, by direct substitution once more

$$ba = a^2b$$

$$\implies aba^2 = a^2b$$

- (ii) Now consider  $a^2bab$ . Note that  $ba = a^2b$  by the given. By substitution, therefore, we obtain

$$a^2bab = a^2a^2bb = a^4b^2$$

Since  $|a| = 3, |b| = 2 \implies a^4 = a, b^2 = e$  by properties of orders of elements. Therefore,

$$a^4b^2 = a$$

$$\implies a^2bab = a$$

□

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<sup>\*</sup>Email: [jzhang72@terpmail.umd.edu](mailto:jzhang72@terpmail.umd.edu)

- 36.** Suppose that  $H$  is a subgroup of  $Z$  under addition and that  $H$  contains  $2^{50}$  and  $3^{50}$ . What are the possibilities for  $H$ ?

*Proof.*  $H$  is a nonempty subset of  $Z$  that immediately contains  $0, 2^{50}$ , and  $3^{50}$ . By the One-Step Subgroup Test,  $H$  is a subgroup if  $\forall a, b \in H, ab^{-1} \in H$ . Additionally, by the Two-Step Subgroup Test,  $H$  is a subgroup if  $\forall a, b, c \in H, ab \in H$  and  $c^{-1} \in H$ . Therefore, all linear combinations of  $2^{50}$  and  $3^{50}$  are also in  $H$ . To formalize this,

$$H = \{x2^{50} + y3^{50} \mid x, y \in \mathbb{Z}\}$$

Considering the trivial prime factor decomposition of  $2^{50}$  and  $3^{50}$ , it is easy to see that they share no prime factors, so their GCD is 1, and they are relatively prime. Therefore, 1 must be in  $H$ , and as a consequence of Bezout's Identity, for any integer  $z \in \mathbb{Z}$ , we can find  $x, y \in \mathbb{Z}$  such that  $z = x2^{50} + y3^{50}$ . Therefore,  $H$  must contain all integers, so  $H$  is equal to  $Z$ .  $\square$

**60.** In the group  $\mathbf{R}^*$  find elements  $a$  and  $b$  such that  $|a| = \infty$ ,  
 $|b| = \infty$  and  $|ab| = 2$ .

*Proof.* Let  $a = 2, b = -\frac{1}{2}$ . Note that  $a$  and  $b$  both have *infinite order* because there does not exist a natural number such that  $a^n = e = 1$  or  $b^n = e = 1$ . Thus,  $|a| = \infty, |b| = \infty$ . Now consider the element  $ab$ .

$$ab = (2)\left(-\frac{1}{2}\right) = -1 \implies (ab)^2 = 1 \implies |ab| = 2$$

as desired. To generalize this, let  $a \neq 1$  be in  $R^*$ . If  $b = -\frac{1}{a} \in R^*$  such that  $ab = -1$ , then  $a, b$  satisfy  $|a| = \infty, |b| = \infty$ , and  $|ab| = 2$ .  $\square$

80. Let  $G$  be a finite Abelian group and let  $a$  and  $b$  belong to  $G$ . Prove that the set  $\langle a, b \rangle = \{a^i b^j \mid i, j \in \mathbb{Z}\}$  is a subgroup of  $G$ . What can you say about  $|\langle a, b \rangle|$  in terms of  $|a|$  and  $|b|$ ?

*Proof.* First let us prove that the set  $\langle a, b \rangle$  is a subgroup of  $H$ . We can prove this by satisfying the conditions for either the One-Step Subgroup Test or the Two-Step Subgroup Test

1. For any  $a, b \in H$ , since  $a^i b^j$  is also in  $H$  for all  $i, j \in \mathbb{Z}$ , if we set  $i = 1, b = -1$ , then  $ab^{-1}$  must be in  $H$ , and so  $H$  must be a subgroup of  $G$ .
2. For any  $a, b \in H$ , since  $a^i b^j$  is also in  $H$  for all  $i, j \in \mathbb{Z}$ ,  $ab$  must also be in  $H$  by setting  $i = 1, j = 1$  and  $a^{-1}$  must also be in  $H$  by setting  $i = -1, b = 0$ . Therefore,  $H$  is a subgroup of  $G$ .

Now let us describe  $|\langle a, b \rangle|$  in terms of  $|a|$  and  $|b|$ . Suppose  $|a| = x, |b| = y$  and  $x, y \in \mathbb{N} + \{0\}$ . We're trying to find the minimum  $n \in \mathbb{N} + \{0\}$  such that

$$|\langle a, b \rangle| = |a^i b^j| = (a^i b^j)^n = a^{in} b^{jn} = e$$

For  $a^{in} b^{jn} = 1$ , we need  $in$  to be a multiple of  $x$  and  $jn$  to be a multiple of  $y$  so that  $a^{in} = e$  and  $b^{jn} = e$ . Therefore, let

$$n_1 = \frac{LCM(i, x)}{i} \qquad n_2 = \frac{LCM(j, y)}{j}$$

Thus we've found  $n_1$  and  $n_2$  such that  $a^{in_1} = e$  and  $b^{jn_2} = e$ . To find the minimum  $n$  such that both of these are true, simultaneously, let

$$n = LCM(n_1, n_2) = LCM\left(\frac{LCM(i, |a|)}{i}, \frac{LCM(j, |b|)}{j}\right)$$

and so we've expressed  $|\langle a, b \rangle|$  in terms of  $|a|$  and  $|b|$ , as desired.  $\square$

**30.** Prove that a group of order 3 must be cyclic.

*Proof.* Since this group has order 3, denote  $G = \{a, b, c\}$ . All groups must have exactly one identity element. Without loss of generality, let element  $c = e$ , so we have  $G = \{a, b, e\}$  and now only have 2 unknown elements. Observe that in groups, for each element, there must exist an inverse. Therefore,  $ab = ba = e \implies a^{-1} = b$ . Finally, observe that  $ab = aa^{-1} = a^0 = e$ . Therefore, we can represent  $G$  as

$$G = \{a, b, c\} = \{a^1, a^{-1}, a^0\} = \langle a \rangle$$

so  $a$  is a generator and  $G$  is cyclic, as desired. □

**58.** Prove that an infinite group must have an infinite number of subgroups.

*Proof.* Let  $G$  be an infinite group, and let  $a$  be any element in  $G$ . We will show that  $\langle a \rangle$  is a subgroup of  $G$ , and therefore, since there are an infinite number of elements in  $G$ , this will show that there are an infinite number of subgroups of  $G$ . Since  $a \in \langle a \rangle$ ,  $\langle a \rangle$  is not empty. Let  $a^n, a^m \in \langle a \rangle, n, m \in \mathbb{Z}$ . Then,

$$a^n(a^m)^{-1} = a^{n-m} = a^n a^{-m} \in \langle a \rangle$$

Therefore, by the One-Step Subgroup Test,  $\langle a \rangle$  is a subgroup of  $G$ . Therefore, there are an infinite number of subgroups of  $G$ , as desired.  $\square$

**72.** Let  $a$  and  $b$  belong to a group. If  $|a| = 12$ ,  $|b| = 22$ , and  $\langle a \rangle \cap \langle b \rangle \neq \{e\}$ , prove that  $a^6 = b^{11}$ .

*Proof.* Let  $a, b \in G$ .  $|a| = 12 \implies a^{12} = e$  and  $|b| = 22 \implies b^{22} = e$ . Since  $\langle a \rangle \cap \langle b \rangle \neq \{e\}$ ,

$$\exists x \in \langle a \rangle \cap \langle b \rangle \text{ s.t. } x = a^i = b^j$$

Note that the order of  $x$  must divide 12 and 22. The divisors of 12 and 22 are 1 and 2. We do not seek 1 because if the order of  $x$  is 1, then  $x = e$ , but we pre-specified that  $x \neq e$ . Therefore, the order of  $x$  must be 2  $\implies |x| = 2 \implies x^2 = e$ . Therefore, since  $x = a^i = b^j$ ,

$$x^2 = a^{2i} = b^{2j} = e \implies i = 6, j = 11$$

because  $a^{12} = e = b^{22}$ . Thus,

$$(a^6)^2 = (b^{11})^2 \implies a^6 = b^{11}$$

as desired. □